

PrimeStudios Printing Website (Node.js)

End-to-End Workflow & Features

1) Roles

- ❖ **Guest (not logged in):** Sees the same storefront as everyone; can browse products, pricing info, samples, and FAQs.
- ❖ **D2C User (Individual):** Fixed pricing, upload own design / pick from templates / hire a designer, regular checkout.
- ❖ **B2B User (Business):** Tiered & city-based pricing, **wallet** for payments, minimum order quantity (MOQ), can upload design or hire a designer, bulk reorders, GST invoices.
- ❖ **Admin (Back Office):** Controls catalogue, pricing rules, wallet ledgers, production stages, shipments, users, designers, and reports.

2) Public Website (same for all before login)

I. Home / Storefront

- A. Product categories (cards, flyers, stickers, tags, stationery, etc.).
- B. “How it works” (3 steps: Design → Approve → Print & Deliver).
- C. Portfolio, materials/finishes, delivery timelines, FAQs.

II. Product Detail Page (PDP)

- A. Options: size, paper, finish, quantity presets.
- B. Price shown (D2C fixed list price by default).
- C. Buttons: **Upload Design**, **Use Template**, **Hire a Designer** (opens intent).

III. Account

- A. Sign up / Login (email/phone OTP).

- B. After login the experience changes based on role (D2C vs B2B), but the visual design stays the same.

3) Account & Onboarding

- A. **D2C**: Basic profile, addresses, phone/email.
- B. **B2B**: Company name, GSTIN, billing & multiple shipping addresses, city/zone, optional credit terms, upload KYC (PAN/GST doc). Admin verifies & assigns **B2B role**, MOQ, price tiers, city pricing zone.

4) Shared Core Flow (applies to both D2C & B2B)

A. Select Product → Configure → Provide Artwork → Approve Proof → Pay → Track → Receive Delivery

Detailed steps:

- 1) **Select & Configure**: Choose product, material, finish, quantity.
- 2) **Design Option**:
 - a) **Upload Artwork** (PDF/AI/CDR/PNG/JPG).
 - b) **Use Templates** (editable online; brand colors, text).
 - c) **Hire a Designer** (creates a design job; see Section 8).
- 3) **Pre-press Checks (Human)**:
 - a) Auto checks: file size, resolution, bleed, colour profile.
 - b) If issues → status “**Artwork Issue**” with notes + resubmit.
- 4) **Digital Proof**: System generates soft proof; user must **Approve** or **Request Revision**.
- 5) **Payment**:
 - a) D2C → standard checkout (UPI / card / netbanking / CoD optional).
 - b) B2B → **Wallet** (deduct from balance), or approved credit if enabled.
- 6) **Production Stages** (live status in profile):

- a) **Order Received → Pre-press → In Production → Quality Check → Packing → Ready to Dispatch → Dispatched (AWB) → Out for Delivery → Delivered.**

7) **Delivery Tracking:**

- a) Courier name + AWB visible; click to track in real time.
- b) ETA shown. Notifications on key milestones (email/SMS/WhatsApp).

8) **Post-delivery:**

- a) Download invoice (GST for B2B).
- b) Rate & review; raise a support ticket if needed.
- c) **Reorder** button.

“Live status updates” after login mirrors the kind of experience Printers Club advertises. printersclub.in

5) D2C (Individual) — Detailed User Flow

1. **Login** (or checkout as guest → prompt to create account).
2. **Pick Product & Options.**
3. **Choose Design Route:**
 - a. Upload your own design, or
 - b. Use templates (customise in browser), or
 - c. Hire a designer (small job for layout/creative).
4. **See Fixed Pricing** (same across cities; shipping calculated at checkout).
5. **Approve Proof → Pay** (UPI/card/netbanking).
6. **Track Order & Delivery** from **My Orders:**
 - a. Visual progress bar of stages.

- b. Courier tracking link and notifications.
- 7. **Receive Order** → **Invoice & Reorder** from order history.

6) B2B (Business) — Detailed User Flow

- ❖ **Login** (B2B role recognised).
- ❖ **Pick Product & Options** (includes bulk quantities).
- ❖ **B2B Pricing Rules Apply Automatically:**
 - **Quantity Tiered Pricing** (e.g., 1–5K, 5–10K, 10K+).
 - **City/Zone-based Pricing** (regional cost differences).
 - **MOQ enforced** (cannot add to cart below MOQ).
- ❖ **Design Options:**
 - Upload artwork (multi-file support per SKU/variant).
 - Hire a designer (complex DTP or creative; job linked to PO).
- ❖ **Purchase using Wallet / Credit:**
 - **Wallet:** Recharge via UPI/card/netbanking; ledger shows debits/credits.
 - Optional **credit line** if approved (Admin sets limit & terms).
- ❖ **Bulk PO & Reorders:**
 - Save configurations as “Quotes” or “PO Drafts”.
 - Quick **Reorder** from past jobs; CSV upload for multi-SKU orders.
- ❖ **Delivery & Multi-address Support:**
 - Split shipments by warehouse or ship-to location.
 - Track per shipment (AWB per package).
- ❖ **Documents:**
 - GST invoice download, e-way bill field, and delivery challans.

- ❖ **Status & Notifications** mirror D2C but with PO/line-level visibility.

7) Order Statuses (shown live in user profile)

- **Order Received**
- **Pre-press Check**
- **Proof Shared**
- **Revision Requested** (customer action pending)
- **Approved for Print** (customer approved)
- **In Production**
- **Quality Check**
- **Packing**
- **Ready to Dispatch**
- **Dispatched** (AWB visible)
- **Out for Delivery**
- **Delivered**
- **On Hold** (issue: file/payment/stock)
- **Cancelled / Returned**

Users see a **progress bar + timestamps**; notifications fire on each change. (Printers Club emphasises live status post login; we'll replicate that clarity. printersclub.in)

8) “Hire a Designer” Flow (D2C & B2B)

- I. User clicks **Hire a Designer** on PDP or Cart.
- II. **Design Brief Form**: brand, text, logo upload, references, deadline.

- III. **Quote & ETA:** auto baseline + human confirmation.
- IV. **Payment:**
 - A. D2C: pay at checkout (adds design fee).
 - B. B2B: deduct from wallet/credit.
- V. **Design Chat & Files:** in-app messaging, versioned files, approvals.
- VI. **Proof Approval:** moves order to **Approved for Print**.
- VII. **Design Ledger:** stored in “My Designs” for reorder.

9) Pricing Logic

- 1) **D2C:** Fixed product pricing (same across cities); standard shipping table or live rates.
- 2) **B2B:** Dynamic pricing engine:
 - a) **Quantity tiers** (per SKU & finish).
 - b) **City/Zone multipliers** (A/B/C zones, or custom city rates).
 - c) **MOQ per product** (hard stop below MOQ).
 - d) **Contract pricing** (optional per customer).
- 3) **Coupons/Promos** (D2C), **Volume Discounts** (B2B).

10) Payments

- 1) **D2C:** UPI, cards, netbanking, wallet gift credits (if any), COD optional.
- 2) **B2B Wallet:** Recharge (UPI/cards/netbanking), auto top-up alerts, low balance warnings, full ledger export (CSV), refunds as **wallet credits** or **credit notes**.
- 3) **Tax invoices** generated at payment capture.

11) Shipping & Tracking

- A. Set **serviceable cities & courier integrations** (Shiprocket/Delhivery/Xpressbees, etc.).
- B. Show delivery ETA on PDP & checkout, then live **AWB tracking** after dispatch.
- C. Partial shipments supported (line item wise).
- D. Pincodes & zone rules configurable by Admin

13) Admin Backend — Key Features

Catalog & Content

- Products, variants, finishes, add-ons, templates.
- Media library, spec sheets sample images.

Pricing & Rules

- D2C price lists (MRP).
- B2B: quantity tiers, city/zone pricing, contract pricing, MOQs.
- Tax (GST slabs), HSN, shipping rules & pincode zones.
- Coupons (D2C), volume discounts (B2B).

Orders & Production

- Order queue with filters.
- File inspection panel (pre-press checklist; approve/flag).
- Move orders across stages; bulk actions.
- Assign to the designer/production line/press.

Wallet & Finance (B2B)

- Wallet top-ups, debits, refunds; ledger & exports.

- Credit limits, dues, reminders, statements.
- GST invoices, credit notes, e-way fields.

Logistics

- Courier booking, label/AWB generation, pickup scheduling.
- Multi-warehouse stock for substrates/consumables (optional).

Users & Permissions

- Role-based access (Admin, Ops, Pre-press, Designer, Finance, Support).
- B2B account management (KYC, zones, price lists).

Design Jobs

- Queue, quotes, proof history, versioning, approvals.

CMS & Pages

- Home, FAQs, policy pages, blog/portfolio.

Reports & Analytics

- Sales (D2C/B2B), product mix, wallet balances, top clients, abandoned carts.

Audit & Compliance

- Activity logs, file access logs, GST-ready invoices, and data retention controls.

14) Tech Stack (proposed, Node.js)

1. **Frontend:** Next.js (React), SSR for SEO + fast PDPs; Tailwind; form validation.
2. **Backend:** Node.js (NestJS/Express); REST/GraphQL APIs.
3. **DB:** PostgreSQL (pricing rules, orders, wallet ledgers); Redis (queues, sessions); Elasticsearch (search, logs) optional.

4. **Storage:** S3-compatible for artwork & proofs; signed URLs.
5. **Queues:** BullMQ/RabbitMQ for pre-press checks, proof generation, notifications.
6. **Integrations:** Payment gateway (Razorpay/PayU), Shipping (Shiprocket/Delhivery), SMS/WhatsApp, Email (SES/SendGrid).
7. **Auth:** JWT + refresh; OTP for phone; RBAC (role-based access control).
8. **Security:** HTTPS, file type/AV scanning, PII minimisation, audit logs, rate limiting, 2FA for Admins.
9. **Observability:** Centralised logs, error tracking, uptime alerts.

15) Data Model (high level)

- 1) **User** (role, KYC, city/zone)
- 2) **Organization** (B2B accounts, credit terms)
- 3) **Product** → **Variant** → **PriceRule** (D2C/B2B tiers, city multipliers, MOQ)
- 4) **Cart/Order** → **OrderItem** → **Files** (artwork, proof)
- 5) **DesignJob** (brief, quotes, versions, messages)
- 6) **WalletAccount** → **WalletTransaction**
- 7) **Shipment** (AWB, courier, events)
- 8) **Ticket** (issues, evidence, resolution)
- 9) **StatusEvent** (timeline for orders)

16) What the Customer Sees (Summary)

- 1) Same site before login.
- 2) After login:

- a) **D2C**: fixed pricing; simple checkout; track order with live stages and courier tracking.
 - b) **B2B**: quantity/city-based pricing; **wallet**; MOQ; PO drafts; multi-address shipping; invoices & statements.
- 3) Both see a **live progress bar** for every order (from pre-press to delivered) and can **track delivery** in real time. Admin updates instantly reflect in customer profiles (web + notifications).

17) Assumptions (to validate with client)

- 1. D2C pricing is **fixed** across cities; B2B pricing varies by **quantity tier** and **city/zone**.
- 2. B2B orders must meet **MOQ** per product.
- 3. Wallet is mandatory for B2B (credit optional).
- 4. Courier integration supports AWB + live tracking links.

18) Next Steps (build plan)

- 1. Finalize product list, finishes, and option trees.
- 2. Lock pricing model: tiers, zones, MOQs.
- 3. Approve order status names & SLA targets.
- 4. Confirm payment + courier partners.
- 5. UX wireframes: PDP, cart, proof screen, tracking page, wallet.
- 6. Admin specs: pricing engine, wallet, production board.